General Pathology & Genetics Study Guide

Focused Review for Questions 1–50 (including Cases A to F)

How to use this guide: Study the yellow "High-Yield" boxes first. They contain the facts most directly tested in Questions 1–50. Then use the question-to-concept map at the end to connect each question with the correct concept.

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1. Cellular Injury, Death, and Inflammation Basics

- **Cellular Death**:
- **Apoptosis**: Programmed, active, organized cell death. The cell shrinks and fragment structures are phagocytized cleanly **without inducing an inflammatory response**.
- **Necrosis**: Pathological, accidental cell death from injury. The cell swells and ruptures, releasing lysosomal enzymes that damage surrounding tissues and **induce acute inflammation**.
- **Phagocytosis Cascade**: The process of engulfing and destroying pathogens or cellular debris.
- *Steps in order*: **Chemotaxis** (cell migration to the site of injury) → **Opsonization** (coating target with proteins for recognition) → **Engulfment** (ingesting target) → **Degradation** (intracellular killing/lysis).
- **The Inflammation Cascade**: A protective tissue response to injury, divided into acute and chronic.
- **Acute Inflammation**: Rapid onset, short duration. Characterized by **fluid exudation (edema)** and the migration of white blood cells, predominantly **neutrophils** (the first line of defense).
- **Chronic Inflammation**: Slow onset, long duration. Involves **macrophages**, lymphocytes, and plasma cells. Macrophages phagocytize debris and release vasoconstrictive chemical mediators that maintain chronic inflammation.
- **Suppuration (Pus)**: Suppuration is the formation of purulent exudate (pus). The primary white blood cells responsible are **neutrophils and macrophages** that ingest bacteria and degenerate at the site of injury.

> **HIGH-YIELD FOR QUESTIONS 1, 2, 3, 4, 5, 8, 9, 31, 32, 34, 37, 38: Cellular Death & Inflammation** * **Apoptosis** is programmed cell death that **does not cause inflammation**. * **Neutrophils** are the first cells to arrive in **acute inflammation**. * **Suppuration** is primarily composed of **neutrophils and macrophages**. * **Macrophages** produce chemical mediators that maintain **chronic inflammation**. * **Gingivitis** is an example of a disease characterized by **chronic inflammation**. * **Leukocytosis** (elevated WBCs) is a typical outcome of inflammation; **leukopenia** (low WBCs) is NOT.

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2. Cardinal Signs & Chemical Mediators of Inflammation

- **Five Cardinal Signs of Inflammation**:
- **Rubor (Redness)**: Caused by vasodilation and **increased vascularity** (blood flow) to the site of injury.
- **Calor (Heat)**: Caused by a combination of increased blood flow and the release of inflammatory mediators.
- **Tumor (Swelling/Edema)**: Caused by the **exudation of fluid** (plasma and proteins) from leaky vessels into the interstitial tissue space.
- **Dolor (Pain)**: Caused by the **stretching of local pain receptors** due to edema, and the release of pain-inducing mediators.

- **Functio laesa (Loss of function)**: Resulting from pain and swelling.
- **Chemical Mediators**: Small molecules that coordinate the inflammatory response.
- **Histamine**: Secreted primarily by **mast cells**. It causes arteriole dilation, increases vascular permeability (edema), and induces smooth muscle contraction.
- **Serotonin**: Acts as a vasoconstrictor and increases vascular permeability during immunologic reactions.
- **Bradykinin**: Increases vascular permeability, contracts smooth muscle, dilates vessels, and triggers **inflammatory pain**.
- **Arachidonic Acid Metabolites**: Precursor to prostaglandins (which cause fever/pain) and leukotrienes.
- **Interleukin-6 (IL-6)**: Moderates susceptibility, development, and progression of **autoimmune and inflammatory diseases**.
- **TNF & IL-1**: Cytokines responsible for endothelial activation, systemic acute-phase reactions (fever), and the hemodynamic effects of septic shock.
- **Nitric Oxide (NO)**: Relaxes vascular smooth muscle causing vasodilation, and acts as a microbicidal agent.
- **Pro-Resolving Mediators**: Lipoxins, resolvins, and protectins. They actively resolve inflammation, enhance microbial clearance, and **reduce pain**.

Mediator	Source	Primary Function	Clinical Relevance
Histamine	Mast Cells	Vasodilation, capillary permeability	Primary mediator in acute inflammation / allergy
Bradykinin	Plasma	Dilation, contracts smooth muscle, pain	Directly stimulates pain receptors (**Dolor**)
IL-6	Macrophages	Cytokine regulation	Drives autoimmune disease progression
Pro-resolving	Lipids	Actively resolves inflammation	Helps **reduce pain** and return cells to homeostasis

HIGH-YIELD FOR QUESTIONS 1, 2, 3, 6, 7, 33, 35, 36: Cardinal Signs & Mediators * **Rubor** (redness) is caused by **increased vascularity**. * **Tumor** (swelling/edema) is caused by **exudation of fluid**. * **Dolor** is responsible for **pain**, caused by stretching pain receptors. * **Histamine** is released by **mast cells** and causes vasodilation and increased permeability. * **IL-6** moderates the progression of **autoimmune and chronic inflammatory diseases**. * **Pro-resolving mediators** help to **reduce pain** and resolve inflammation.

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3. Wound Healing, Repair, and Tissue Regeneration

- **Wound Healing Phases**:
- **Step 1: Hemostasis**: Immediate **thrombus (clot) formation** at the injury site to prevent blood loss.
- **Step 2: Inflammation**: Emigration of neutrophils and macrophages to clear pathogens.
- **Step 3: Proliferation**: Cell migration and division. Epithelial cells migrate and form a **basement membrane under the protective scab within 24 to 48 hours**.

- **Step 4: Granulation Tissue**: Tissue that fills the wound space. It is characterized by being **pink, soft, and containing fragile, leaky new blood vessels** (making it edematous), alongside proliferating fibroblasts.
- **Step 5: Remodeling**: Tissue gains strength. Tissues recover **70-80% of original tensile strength over a 3-month period** of healing as compared to intact skin (it rarely recovers 100%).
- **Types of Healing Intention**:
- **Primary Intention**: Occurs when wound edges are clean and closely approximated (e.g., sutured surgical incision, or narrow drained abscess fistula). Healing occurs with minimal scar.
- **Secondary Intention**: Occurs in larger wounds with separated edges. The gap is filled by abundant granulation tissue, leaving a large scar and undergoing significant wound contraction.
- **Tertiary Intention**: Delayed primary closure. Contaminated wounds are left open for repeated debridement and antibiotic therapy before surgical closure.
- **Healing Anomalies**:
- **Wound Dehiscence**: Separation or splitting open of wound margins due to **deficient scar formation**.
- **Keloid**: A raised, overgrown scar that extends beyond the original wound margins due to **excessive scar growth**.
- **Contracture**: An exaggeration of normal contraction during healing, resulting in physical deformity.
- **Fibrosis**: Pathological deposition of excessive collagen that sustains growth factors, causing massive scarring and dysfunction.
- **Factors Influencing Healing**:
- **Local Factors**: Inhibit healing at the wound site (e.g., **size and location of the wound, foreign material**, infection).
- **Systemic Factors**: Body-wide factors (e.g., **blood supply/ischemia**, diabetes, nutrition, age, steroid use).

HIGH-YIELD FOR QUESTIONS 10, 11, 12, 13, 14, 15, 16, 17, 39, 40, 41: Wound Healing & Repair * **Thrombus/clot formation** is the **first step** in wound healing. * Epithelial cells form a **basement membrane** under the scab in **24-48 hours**. * **Granulation tissue** is pink and soft, and contains **fragile, leaky vessels**. * Wounds recover **70-80% of tensile strength** by **3 months**. * **Blood supply** is a **systemic factor** in healing; **wound size & foreign material** are **local factors**. * **Wound Dehiscence** is caused by **deficient scar formation**. * **Fibrosis** is the extensive deposition of collagen. * Compromised blood supply (severe ischemia) can result in tissue necrosis and **amputation**.

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4. Stem Cells & Regenerative Medicine

- **Basic Properties**: Stem cells are characterized by **self-renewal** (the capacity to undergo division or remain **quiescent/resting**) and **potency** (the ability to differentiate into other cell types).
- **Stem Cell Sources**:
- **Autogenous**: Stem cells harvested from the same individual being treated (no rejection risk).
- **Allogenous**: Stem cells harvested from another individual of the same species.
- **Levels of Potency**:
- **Totipotent**: Can differentiate into any cell type, including extraembryonic membranes (e.g., zygote).
- **Pluripotent**: Can differentiate into all cells of the body, but not extraembryonic tissues (e.g., **embryonic stem cells**, isolated from the **inner cell mass of blastocysts**).
- **Multipotent**: Can differentiate into multiple cell types within a specific lineage (e.g., **adult stem cells**, found in **skin, gut lining, cornea, brain, and bone marrow**).

- **Unipotent**: Can give rise to only one mature cell type.
- **Adult & Dental Stem Cells**:
- The most common commercially available source of adult stem cells is **bone marrow**.
- **Dental Stem Cells**: High-potency adult stem cells found in dental soft tissues. Sources include the **dental pulp, periodontal ligament (PDL), unerupted third molars, and exfoliated primary teeth**. (Note: the mineralized *pulp chamber walls* composed of dentin do not contain stem cells).

> **HIGH-YIELD FOR QUESTIONS 42, 43, 44, 45, 46: Stem Cells** * **Allogenous** stem cells come from **another individual**. * **Embryonic stem cells** are **pluripotent** and isolated from the **inner cell mass of blastocysts**. * **Adult stem cells** are found in **skin, gut lining, and bone marrow**; **bone marrow** is the most common commercial source. * **Dental stem cells** are found in **dental pulp, PDL, third molars, and primary teeth**; the **pulp chamber wall (dentin)** is NOT a source.

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5. Genetics and Patterns of Inheritance

- **The Human Genome**: Humans possess approximately **20,000 to 25,000 protein-coding genes**.
- **DNA Code**: The genetic code is written using four nitrogenous base letters: **Adenine (A), Thymine (T), Cytosine (C), and Guanine (G)**.
- **Modes of Inheritance & Transmission Risks**:
- **Autosomal Dominant**: A single copy of the mutated gene causes the disorder. A carrier/affected parent has a **50% chance of passing the trait** to each offspring. Examples: Huntington's disease, von Willebrand disease, Marfan syndrome.
- **Autosomal Recessive**: Two copies of the mutated gene are required. Two carrier parents (Aa x Aa) have a **25% chance of having an affected child** (aa) with each pregnancy. Examples: Sickle cell anemia, cystic fibrosis, phenylketonuria (PKU).
- **X-Linked Recessive**: Gene located on the X chromosome. Primarily affects males (who only have one X). There is **no male-to-male transmission**. Affected males transmit the mutated X to **all daughters**, making them obligate carriers. Example: **Duchenne muscular dystrophy**.
- **Multifactorial Inheritance**: Disorders caused by the combined effects of multiple genes and environmental factors. Examples: **Cleft lip and/or cleft palate, coronary heart disease**.
- **Prenatal Diagnostics**: Fetal genetic testing is performed on cells obtained via **amniocentesis, chorionic villus biopsy (CVS), or umbilical cord blood sampling**.

> **HIGH-YIELD FOR QUESTIONS 18, 19, 20, 21, 22, 23, 24, 25, 47, 48, 49, 50: Genetics & Inheritance** * The human genome has **20,000 to 25,000 genes**. * The four DNA letters are **Adenine, Thymine, Cytosine, and Guanine**. * **Autosomal dominant** disorders have a **50% transmission risk**. * **Autosomal recessive** carrier parents have a **25% risk** of an affected child. * **Duchenne muscular dystrophy** is an **X-linked recessive** disorder. All daughters of affected males are carriers. * **Cleft lip and/or palate** and **coronary heart disease** are **multifactorial** disorders. * Indication for genetic counseling: **Family history of cleft lip/palate**. * Fetal genetic cells are gathered via **amniocentesis, chorionic villus biopsy, and umbilical cord blood**.

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6. Diagnostic Process and Clinical Assessment

- **Clinical Evaluation Techniques**:

- **Visual Examination**: Assessing the location, shape, size, color, delineation of borders, and texture of a lesion.
- **Palpation**: Feeling a lesion to determine compressibility, tenderness, and color changes under pressure (e.g. blanching).
- **Auscultation**: Listening to sounds (e.g., popping/clicking TMJ).
- **Probing**: Evaluating tissue defects or measuring pocket depths.
- **Aspiration**: Withdrawing fluid (e.g., pus from abscesses, cyst fluid).
- **Diagnostic History Categories**:
- **Habits**: Alcohol use, tobacco use, and oral behaviors (e.g., nail-biting, cheek-biting).
- **Demographics**: Patient age, gender, and race/ethnicity.
- **Recent History**: History of local injury, infection, or surgery.
- **Awareness of Condition**: Patient's reports of pain, duration, and response to stress or foods.
- **Working Diagnosis Classifications**: The initial list of possibilities is a **working diagnosis**, which includes **differential diagnosis, preliminary diagnosis, and tentative diagnosis**. (Note: "probable diagnosis" is not a recognized clinical term).

> **HIGH-YIELD FOR QUESTIONS 26, 27, 28, 29, 30: Diagnostic Process** * A mucosal swelling is classified as an **enlargement of soft tissues**. * **A solid soft tissue mass cannot be probed**; it is assessed via palpation, aspiration, and visual inspection. * **Tobacco/alcohol use** are classified as patient **habits**. * **Pain and duration** fall under patient **awareness of condition**. * **Probable diagnosis** is a fabricated term, NOT a form of working diagnosis.

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7. Case Study Reviews (Cases A to F)

- **Case A Review (Questions 1 to 4)**: Steve Perry presents with generalized gingivitis, red fiery tissues, heavy biofilm, and bleeding/pain during flossing.
- *Redness* is caused by **increased vascularity**.
- *Edema* is caused by **exudation of fluid**.
- *Pain* is mediated by **dolor** (stretching pain receptors).
- *Reaction* is a **vascular and cellular reaction** in the inflammatory process.
- **Case B Review (Questions 5 to 10)**: Evelyn Goddard presents with a periodontal abscess (buccal mucosa of #30), swelling, redness, pain, and a fistula expressing pus.
- *Abscess response* represents **acute inflammation** (neutrophils and fluid exudation).
- *Redness* is known as **rubor**.
- *Edema* is caused by **exudation of fluid**.
- *Pus* consists of **neutrophils and macrophages**.
- *Healing* of the narrow drained fistula will occur by **primary intention**.
- **Case C Review (Questions 11 to 17)**: Sam Adams underwent extraction of impacted tooth #32.
- *First step in healing* is **thrombus/clot formation**.
- *Basement membrane* forms under the scab within **24-48 hours**.
- *Granulation tissue* is pink, soft, and has **immature, leaky vessels** (exception: they are not leak-free).
- *Tensile strength* recovers to **70-80% over 3 months**.

- *Systemic factor prolonging healing* is **blood supply** (wound size/location are local).
- **Case D Review (Questions 18 to 21)**: Monica Salt is a pregnant patient concerned about family history of cleft lip and palate.
- *Family history* is a primary reason for **genetic counseling**.
- *Gene mutations* causing hereditary diseases are mostly **submicroscopic**.
- *Genetic material* of the fetus is obtained via **amniocentesis, chorionic villus biopsy, and umbilical cord blood**.
- *Cleft lip/palate* is inherited via **multiple factorial inheritance**.
- **Case E Review (Questions 22 to 25)**: A 4-year-old boy has Duchenne muscular dystrophy.
- *Muscular dystrophy* is an **X-linked recessive disorder**.
- *Transmission*: Affected males transmit the mutant X to **all daughters** (making them carriers). There is **no male-to-male transmission**.
- *Multifactorial inheritance example* is **coronary heart disease** (muscular dystrophy is X-linked).
- *Genetic material analysis* is done via **karyotyping, chromosome analysis, and PCR**.
- **Case F Review (Questions 26 to 30)**: Janis Johnson has a painless, normal-colored buccal mucosa mass that she bites occasionally, causing ulceration.
- *Primary manifestation* of this mass is **enlargement of soft tissues**.
- *Examination limitation*: **A mass cannot be probed**.
- *Habits*: Her cheek-biting, along with tobacco/alcohol use, represent patient **habits**.
- *Awareness of condition*: Pain and duration represent patient **awareness**.

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8. Question-to-Concept Map

Question	Case	Topic	Key Fact to Recall
1	Case A	Inflammation (Redness)	Rubor (redness) in gingivitis is caused by **increased vascularity**.
2	Case A	Inflammation (Edema)	Swelling in gingivitis is caused by the **exudation of fluid**.
3	Case A	Inflammation (Pain)	**Dolor** represents pain, caused by stretching pain receptors.
4	Case A	Inflammation Reaction	The inflammatory process is a **vascular and cellular reaction**.
5	Case B	Acute Inflammation	Abscess response is **acute inflammation** (rapid onset, neutrophil-rich).

6	Case B	Cardinal Signs	**Rubor** is the clinical term for redness.
7	Case B	Edema Process	Edema is caused by the **exudation of fluid** into tissues.
8	Case B	Suppuration Cells	**Neutrophils and macrophages** are the cells forming pus.
9	Case B	Neutrophils	Neutrophils phagocytize and also **contribute to the repair process**.
10	Case B	Intention Healing	Narrow, closely approximated wounds heal via **primary intention**.
11	Case C	Wound Healing Step 1	The first stage of wound healing is **thrombus (clot) formation**.
12	Case C	Epithelialization	The epithelial basement membrane forms under a scab in **24-48 hours**.
13	Case C	Granulation Tissue	Early granulation vessels are immature and **leak fluid (edema)**.
14	Case C	Tensile Strength	Healed wounds recover **70-80% of original strength** at **3 months**.
15	Case C	Healing Factors	**Blood supply** is a systemic factor; wound size/location are local.
16	Case C	Deficient Scarring	**Wound Dehiscence** is wound splitting due to deficient scar tissue.
17	Case C	Healing Factors	**Blood supply** is a systemic factor, not a local factor.
18	Case D	Genetic Counseling	Family history of cleft lip/palate is an indication for **genetic counseling**.

19	Case D	Hereditary Mutations	Most hereditary mutations are **submicroscopic gene mutations**.
20	Case D	Prenatal Testing	Fetal cells are gathered via **amniocentesis, CVS, and cordocentesis**.
21	Case D	Cleft Lip/Palate	Cleft lip and palate is inherited via **multiple factorial inheritance**.
22	Case E	Genetic Disease Type	Duchenne muscular dystrophy is an **X-linked recessive** disorder.
23	Case E	X-linked Inheritance	Carrier females pass the gene to daughters (carriers) and sons (50% affected).
24	Case E	Multifactorial Disorders	**Coronary heart disease** is an example of a multifactorial disorder.
25	Case E	Genetic Evaluation	Genetic material is analyzed using **karyotyping, chromosome analysis, and PCR**.
26	Case F	Mass Classification	A localized mucosal swelling is classified as an **enlargement of soft tissues**.
27	Case F	Diagnostic Techniques	Probing cannot be performed on a solid soft tissue mass.
28	Case F	Diagnostic History	Plaque, alcohol, and tobacco use fall under patient **habits**.
29	Case F	Subjective History	Pain, duration, and food triggers fall under **awareness of condition**.
30	Case F	Working Diagnosis	**Probable diagnosis** is a fabricated term, not a working diagnosis.
31	None	Macrophages	**Macrophages** maintain chronic inflammation by producing mediators.

32	None	Eosinophils	**Eosinophils** defend against parasites and participate in allergic reactions.
33	None	Histamine	**Histamine** causes vasodilation, permeability, and smooth muscle contraction.
34	None	Apoptosis	Programmed cell death during phagocytosis is **apoptosis**.
35	None	Resolving Mediators	Pro-resolving lipid mediators **reduce pain** and resolve inflammation.
36	None	Cytokines	**IL-6** moderates the progression of chronic autoimmune diseases.
37	None	Inflammation Outcomes	**Leukopenia** is not an outcome of acute inflammation.
38	None	Chronic Inflammation	**Gingivitis** is a chronic inflammatory oral disease.
39	None	Tissue Repair	**Granulation tissue** is the diagnostic hallmark of wound healing.
40	None	Fibrosis	**Fibrosis** is pathologic scar formation from excessive collagen.
41	None	Tissue Necrosis	Severely compromised blood supply can lead to gangrene and **amputation**.
42	None	Allogenous	**Allogenous** cells are harvested from a different donor.
43	None	Stem Cell Potency	Embryonic stem cells are pluripotent; adult stems are multipotent.
44	None	Stem Cell Commercial source	**Bone marrow** is the most common commercial source of adult stem cells.

45	None	Stem Cell Division	Stem cells undergo self-renewal and can remain **quiescent (resting)**.
46	None	Dental Stem Cells	Dental pulp, PDL, and primary teeth contain stem cells; **dentin walls** do not.
47	None	Human Genome	Humans have approximately **20,000 to 25,000 genes**.
48	None	DNA Structure	DNA bases consist of **Adenine, Thymine, Cytosine, and Guanine**.
49	None	Autosomal Dominant	A parent has a **50% chance** of passing an autosomal dominant trait.
50	None	Autosomal Recessive	Two carrier parents have a **25% chance** of an affected child.

9. Rapid Recall

- **Apoptosis** → Programmed cell death that does not induce inflammation.
- **Neutrophils** → First white blood cells to arrive in acute inflammation.
- **Suppuration (Pus)** → Primarily composed of neutrophils and macrophages.
- **Rubor** → Redness (caused by increased vascularity).
- **Tumor** → Swelling/edema (caused by fluid exudation).
- **Dolor** → Pain (caused by stretching of pain receptors).
- **Calor** → Heat (caused by increased blood flow and mediators).
- **Functio laesa** → Loss of function.
- **Histamine** → Released by mast cells; causes vasodilation and vascular leakage.
- **Thrombus** → Clot formation; first step in wound healing.
- **Granulation Tissue** → Pink, soft, containing fragile, leaky new capillaries.
- **Wound Dehiscence** → Wound separation from deficient scar formation.
- **Allogenous** → Cells derived from a different individual of the same species.
- **Bone Marrow** → Most common commercial source of adult stem cells.
- **Duchenne Muscular Dystrophy** → X-linked recessive disorder (no male-to-male transmission).
- **Cleft Lip/Palate** → Multifactorial inheritance disorder (genetic + environment).
- **Amniocentesis / CVS** → Common methods for obtaining fetal cells for genetic testing.
- **Differential Diagnosis** → Form of working diagnosis.
- **Probable Diagnosis** → A fabricated term (not a working diagnosis).

- **20,000 to 25,000** → Approximate number of genes in the human genome.
- **Adenine, Thymine, Cytosine, Guanine** → Nitrogenous bases of DNA.
- **50%** → Transmission risk of an autosomal dominant disorder to offspring.
- **25%** → Risk of affected child with autosomal recessive carrier parents.